

Chapter 2

Intelligent Agents

We begin by examining agents, environments, and the coupling between them. The observation that some agents behave better than others leads naturally to the idea of a rational agent—one that behaves as well as possible. How well an agent can behave depends on the nature of the environment; some environments are more difficult than others. We give a crude categorization of environments and show how properties of an environment influence the design of suitable agents for that environment.

Agents TF--->'19-20

- An **agent** is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

Human agent:

eyes, ears, and other organs for sensors;
hands, legs, mouth, and other body parts for actuators

Robotic agent:

cameras and infrared range finders for sensors; various motors for actuators

A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.

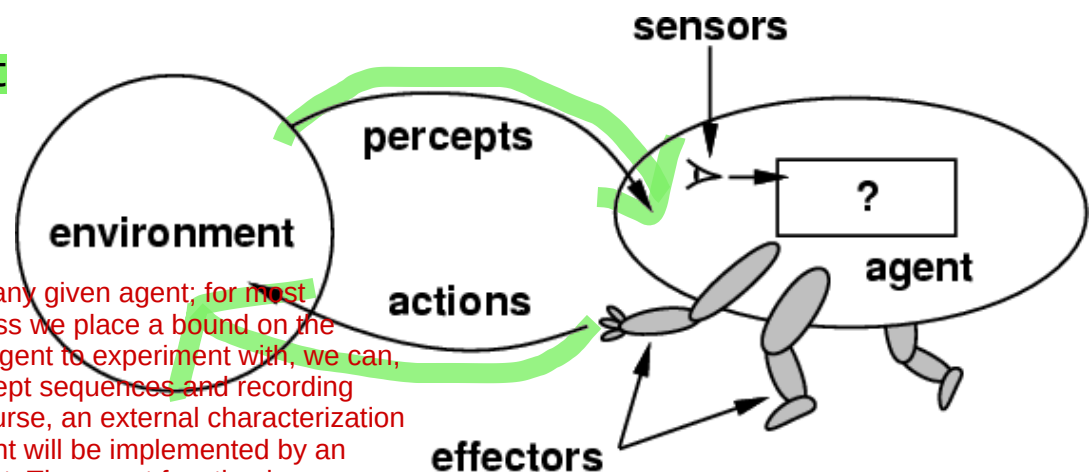
We use the term percept to refer to the agent's perceptual inputs at any given instant. An agent's percept sequence is the complete history of everything the agent has ever perceived. In general, an agent's choice of action at any given instant can depend on the entire percept sequence observed to date, but not on anything it hasn't perceived. By specifying the agent's choice of action for every possible percept sequence, we have said more or less everything there is to say about the agent. Mathematically speaking, we say that an agent's behavior is described by the agent function that maps any given percept sequence to an action.

How to design an intelligent agent?

- An **intelligent agent** perceives its environment via **sensors** and **acts** rationally upon that environment with its **effectors**.
- A discrete agent receives **percepts** one at a time, and maps this percept sequence to a sequence of discrete **actions**.

- **Properties**

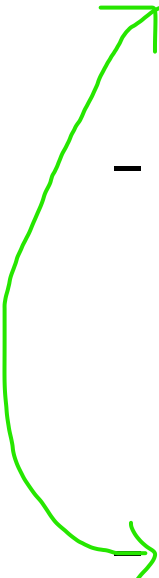
1. **Autonomous**
2. **Reactive to the environment**
3. **Pro-active** (goal-directed)
4. **Interacts with other agents**
via the environment



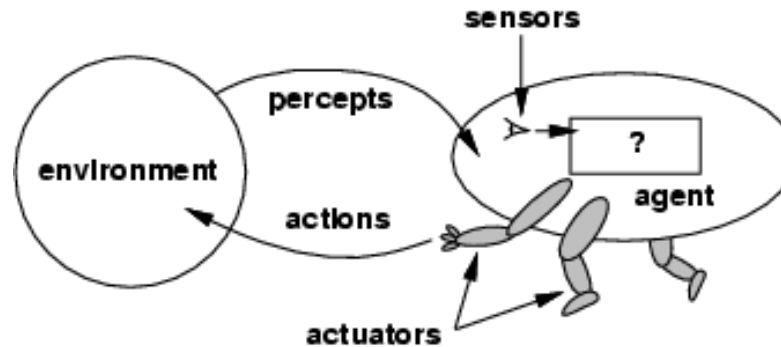
We can imagine tabulating the agent function that describes any given agent; for most agents, this would be a very large table—infinite, in fact, unless we place a bound on the length of percept sequences we want to consider. Given an agent to experiment with, we can, in principle, construct this table by trying out all possible percept sequences and recording which actions the agent does in response. The table is, of course, an external characterization of the agent. Internally, the agent function for an artificial agent will be implemented by an agent program. It is important to keep these two ideas distinct. The agent function is an abstract mathematical description; the agent program is a concrete implementation, running within some physical system.

Sensors/percepts and effectors/actions?

- Humans

- **Sensors**: Eyes (vision), ears (hearing), skin (touch), tongue (gustation), nose (olfaction), neuromuscular system (proprioception)
 - **Percepts**:
 - At the **lowest level** – **electrical signals** from these sensors
 - **After preprocessing** – **objects in the visual field** (location, textures, colors, ...), auditory streams (pitch, loudness, direction), ...
 - **Effectors**: limbs, digits, eyes, tongue, ...
 - **Actions**: **lift a finger**, **turn left**, **walk**, **run**, carry an object, ...
- 

Agents and environments



Agent Function: This refers to a mathematical function f that defines the behavior of an AI agent. It takes as input a history of percepts (observations from the environment) and maps these to an action.

Percept Histories (P):* These are sequences of percepts that the agent has received over time. The asterisk (*) denotes a sequence, indicating that the agent considers not just the current percept, but also previous percepts.

Actions (A): These are the possible actions the agent can take in response to the

- The **agent function** maps from percept histories to actions:

$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$

percepts. The agent function determines which action to take based on the percept history.

- The agent program runs on the physical architecture to produce f

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- agent = architecture + program**

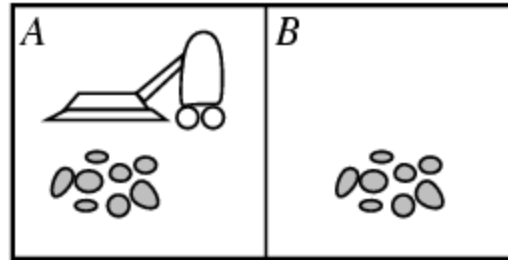
Agent: In AI, an agent is an entity that perceives its environment through sensors and acts upon that environment through actuators.

Architecture: This includes the physical hardware and possibly lower-level software that supports the agent program.

Program: This is the higher-level software (the agent program) that defines the agent's behavior.

Sum: The complete agent is the combination of its architecture (hardware) and program (software). Both components are essential for the agent to function.

Vacuum-cleaner world



- Percepts: location and state of the environment, e.g., [A,Dirty], [A,Clean], [B,Dirty]
- Actions: *Left, Right, Suck, NoOp*

To illustrate these ideas, we use a very simple example—the vacuum-cleaner world . This world is so simple that we can describe everything that happens; it's also a made-up world, so we can invent many variations. This particular world has just two locations: squares A and B. The vacuum agent perceives which square it is in and whether there is dirt in the square. It can choose to move left, move right, suck up the dirt, or do nothing. One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square.

Rational agents

- **Performance measure**: An objective criterion for success of an agent's behavior, e.g.,

- Robot driver?
- Chess-playing program?
- Spam email classifier?

A rational agent is one that does the right thing—conceptually speaking, every entry in the table for the agent function is filled out correctly. Obviously, doing the right thing is better than doing the wrong thing.

- **Rational Agent**: selects actions that is *expected* to maximize its performance measure,

- given percept sequence
- given agent's built-in knowledge

- sidepoint: how to maximize expected future performance, given only historical data

What is rational at any given time depends on four things:

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Rational agents

- Rational Agent → Always try to maximize performance.
- No Agent is Omniscience. Rationality is distinct from omniscience (all-knowing with infinite knowledge)

An omniscient agent knows the actual outcome of its actions and can act accordingly; but omniscience is impossible in reality.

- Agents can perform actions in order to modify future percepts so as to obtain useful information (information gathering, exploration)
- An agent is autonomous if its behavior is determined by its own percepts & experience (with ability to learn and adapt) without depending solely on built-in knowledge
- To survive, agents must have:
 - Enough built-in knowledge to survive.
 - The ability to learn

Task Environment

- Before we design an intelligent agent, we must specify its “task environment”:
we must think about task environments, which are essentially the “problems” to which rational agents are the “solutions”

PEAS:

Performance measure

Environment

Actuators

Sensors

DARPA Robotics Challenge

- A prize competition funded by the US Defense Advanced Research Projects Agency, held from 2012 to 2015
- Aimed to develop semi-autonomous ground robots that could do complex tasks in dangerous, degraded, human-engineered environments

DARPA Robotics Challenge

The initial task requirements for robot entries are

- Drive a utility vehicle at the site
- Travel dismounted across rubble
- Remove debris blocking an entryway
- Open a door and enter a building
- Climb an industrial ladder and traverse an industrial walkway
- Use a tool to break through a concrete panel
- Locate and close a valve near a leaking pipe
- Connect a fire hose to a standpipe and turn on a valve

PEAS

- Example: Agent = robot driver in DARPA Challenge
 - Performance measure:
 - Time to complete course
 - Environment:
 - Roads, other traffic, obstacles
 - Actuators:
 - Steering wheel, accelerator, brake, signal, horn
 - Sensors:
 - Optical cameras, lasers, sonar, accelerometer, speedometer, GPS, odometer, engine sensors,

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Agent = Interactive English Tutor
Performance Measure : Student's score on test
Environment : Set of Students, testing agency
Actuators : Display of exercises, suggestions, corrections
Sensors : Keyboard entry

Agent = Bidding on an item at an auction
Performance Measure: Winning the bid at the lowest possible price while maximizing profit or value and staying within budget.
Environment: Auction house or online auction platform, competitors, auctioneer, auction rules, time constraints.
Actuators: Bidding system interface (buttons, mouse clicks, keyboard inputs), bid increments, bid placement, budget management tools, real-time bidding adjustments.
Sensors: Auction timer, current bid amount, bid history, competing bids, bid alerts, auctioneer announcements, financial limits, competing bid trends.

- Example: Agent = Medical diagnosis system

Performance measure:

Healthy patient, minimize costs, lawsuits

Environment:

Patient, hospital, staff

Actuators:

Screen display (questions, tests, diagnoses, treatments, referrals)

Sensors:

Keyboard (entry of symptoms, findings, patient's answers)

Environment types

- **Fully observable** (vs. **partially observable**):
 - An agent's sensors give it access to the complete state of the environment at each point in time.
- **Deterministic** (vs. **stochastic**):
 - The next state of the environment is completely determined by the current state and the action executed by the agent.
 - If the environment is deterministic except for the actions of other agents, then the environment is **strategic**
 - Deterministic environments can appear stochastic to an agent (e.g., when only partially observable)
- **Episodic** (vs. **sequential**):
 - An agent's action is divided into atomic episodes. Decisions do not depend on previous decisions/actions.

Environment types

- **Static** (vs. **dynamic**):
 - The **environment is unchanged** while an agent is deliberating.
 - The environment is **semidynamic** if the **environment itself does not change** with the passage of time **but the agent's performance score does**
- **Discrete** (vs. **continuous**):
 - A **discrete set** of distinct, clearly defined **percepts and actions**.
 - How we **represent** or **abstract** or **model** the world
- **Single agent** (vs. **multi-agent**):
 - An **agent operating by itself** in an environment. Does the other agent interfere with my performance measure?

Characteristics of environments

	Fully observable?	Deterministic	Episodic	Static	Discrete?	Single agent?
Solitaire						
Driving						
Internet shopping						
Medical diagnosis						

Characteristics of environments

	Fully observable?	Deterministic?	Episodic?	Static?	Discrete?	Single agent?
Solitaire	No	Yes	Yes	Yes	Yes	Yes
Driving	No	No	No	No	No	No
Internet shopping						
Medical diagnosis						

Characteristics of environments

	Fully observable?	Deterministic?	Episodic?	Static?	Discrete?	Single agent?
Solitaire	No	Yes	Yes	Yes	Yes	Yes
Driving	No	No	No	No	No	No
Internet shopping	No	No	No	No	Yes	No
Medical diagnosis						

We have listed the medical-diagnosis task as single-agent because the disease process in a patient is not profitably modeled as an agent; but a medical-diagnosis system might also have to deal with recalcitrant patients and skeptical staff, so the environment could have a multiagent aspect.

Characteristics of environments



	Fully observable?	Deterministic?	Episodic?	Static?	Discrete?	Single agent?
Solitaire	No	Yes	Yes	Yes	Yes	Yes
Driving	No	No	No	No	No	No
Internet shopping	No	No	No	No	Yes	No
Medical diagnosis	No	No	No	No	No	Yes

→ Lots of real-world domains fall into the hardest case!



task environm.	observabl e	deterministic / stochastic	episodic/ sequential	static/ dynamic	discrete/ continuous	agents
crossword puzzle	fully	determ.	sequential	static	discrete	single
chess with clock	fully	strategic	sequential	semi	discrete	multi
poker	Partial	Stochastic	Sequential	Static	Discrete	multi
taxi driving	partial	stochastic	sequential	dynamic	continuous	multi
medical diagnosis	Partial	Stochastic	Sequential	Dynamic	Continuous	Single
image analysis	fully	determ.	episodic	semi	continuous	single
partpicking robot	partial	stochastic	episodic	dynamic	continuous	single
refinery controller	partial	stochastic	sequential	dynamic	continuous	single
interact. tutor	partial	stochastic	sequential	dynamic	discrete	multi

What is the environment for the DARPA Challenge?

- Agent = robotic vehicle
- Environment = 130-mile route through desert
 - Observable? Partially Observable
 - Deterministic? Stochastic
 - Episodic? Episodic
 - Static? Dynamic
 - Discrete? Discrete or Continuous
 - Agents? Multi agents

The DARPA (Defense Advanced Research Projects Agency) Challenge environments can vary depending on the specific challenge, but they often share certain characteristics. Here's an analysis based on typical DARPA challenges:

Observable:

Often, the environment is partially observable. Agents might not have access to the full state of the environment at all times and need to infer missing information from observations.

Deterministic:

The environment can be non-deterministic. Actions may not always lead to the same outcomes due to various uncertainties, such as sensor noise or unpredictable interactions with the environment.

Episodic:

The environment is usually episodic. Challenges are often divided into episodes or tasks with a clear start and end, allowing agents to reset and start fresh for each episode.

Static:

The environment is typically dynamic. It can change over time, often in response to the actions of the agent or due to other dynamic elements present in the environment.

Discrete:

The environment can be continuous or discrete. Depending on the challenge, the state space and action space can either be discrete (e.g., a grid world) or continuous (e.g., navigating through a physical space).

Agents:

There are usually multiple agents. In many challenges, there are multiple agents, either cooperating or competing, which can add to the complexity of the environment

Agent types

- Five basic types in order of increasing generality:

- Table Driven agent

- Simple reflex agents

- Model-based reflex agents

TF--->'20-21

- Goal-based agents

- Problem-solving agents

- Utility-based agents

- Can distinguish between different goals

- Learning agents

Some agent types

- **Table-driven agents**

- use a **percept sequence/action table** in memory to find the next action. They are **implemented by** a (large) **lookup table**. It is not autonomous.

- **Simple reflex agents**

- are **based** on **condition-action rules**, implemented with an appropriate production system. They are stateless devices which do not have memory of past world states. It can not save history.

- **Agents with memory**

- have **internal state**, which is used to keep track of past states of the world.

- **Agents with goals**

- are agents that, in addition to state information, **have goal information** that **describes desirable situations**. Agents of this kind take future events into consideration. **Never thinks about cost.**

- **Utility-based agents**

- base their decisions on **classic axiomatic utility theory** in order to act rationally. **Always thinks about cost.**

Table-driven/reflex agent Architecture

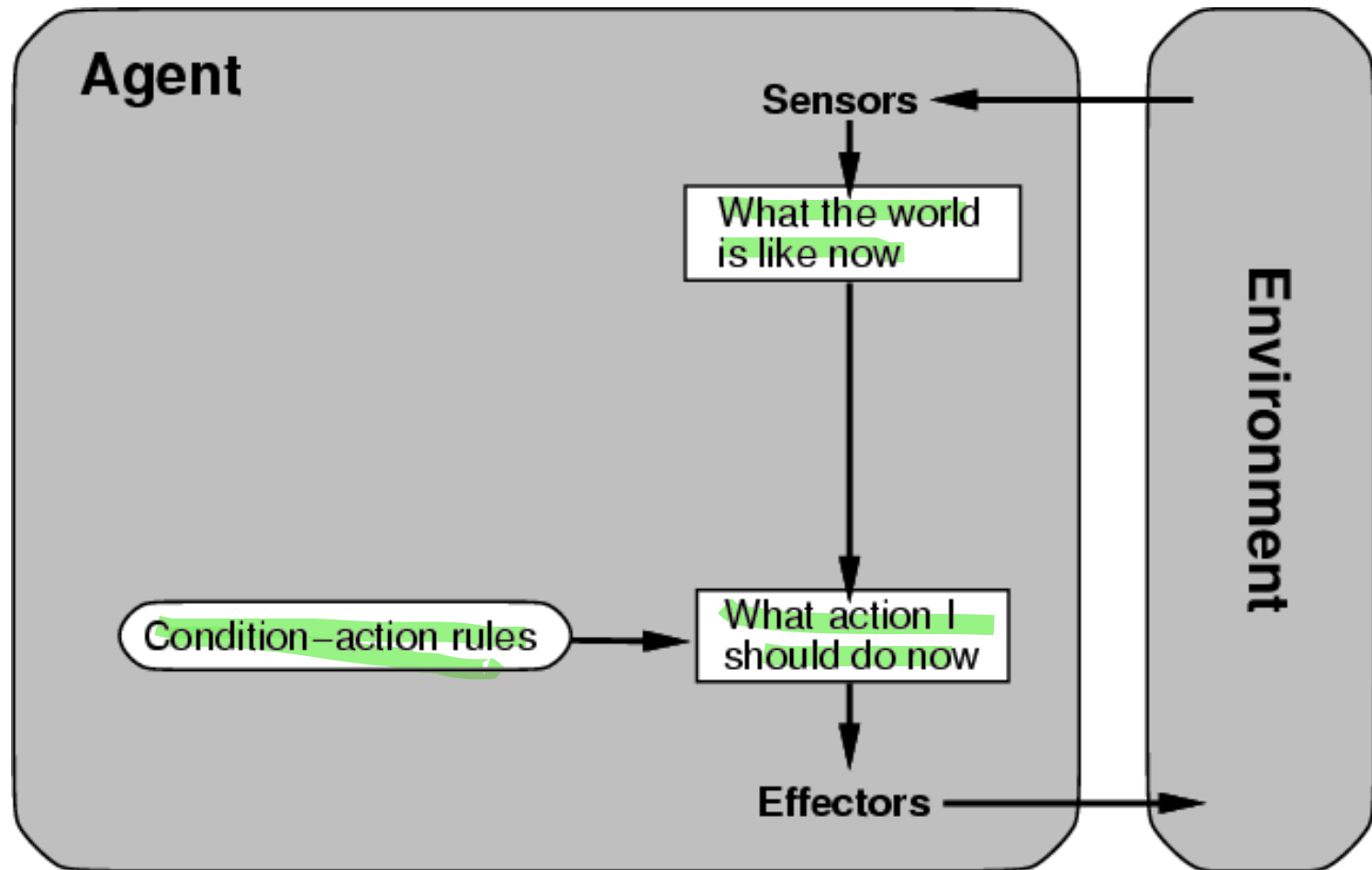


Table-driven agents

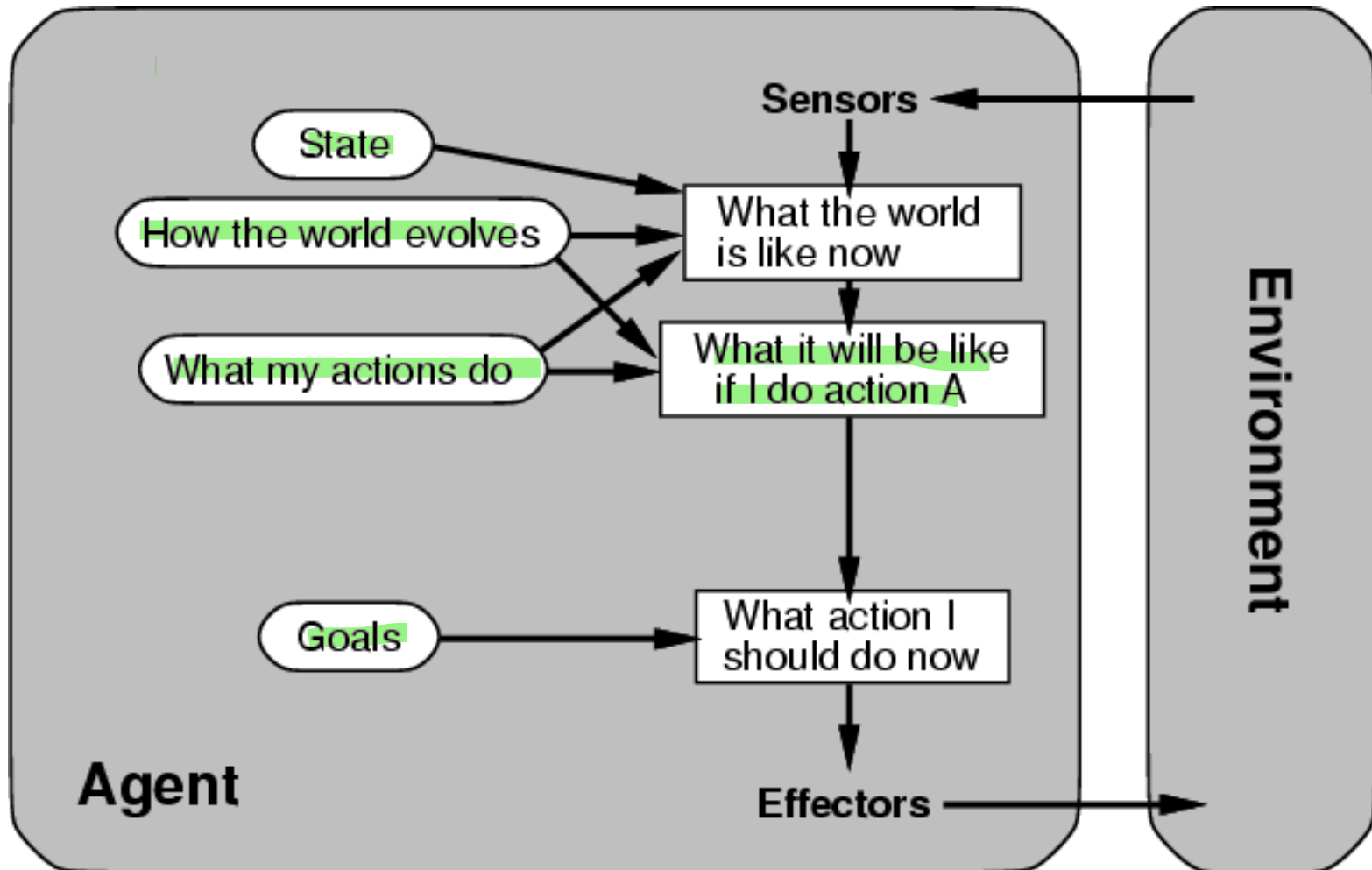
- **Table lookup** of percept-action pairs mapping from every possible perceived state to the optimal action for that state
- **Problems**
 - Too big to generate and to store (Chess has about 10^{120} states, for example)
 - No knowledge of non-perceptual parts of the current state
 - Not adaptive to changes in the environment; requires entire table to be updated if changes occur
 - Looping: Can't make actions conditional on previous actions/states

Simple reflex agents

- **Rule-based reasoning** to map from percepts to optimal action; each rule handles a collection of perceived states
- **Problems**
 - Still usually **too big to generate and to store**
 - Still **not adaptive** to changes in the environment; requires collection of rules to be updated if changes occur
 - Still **can't make actions conditional on previous state**

Goal-based agent: Architecture

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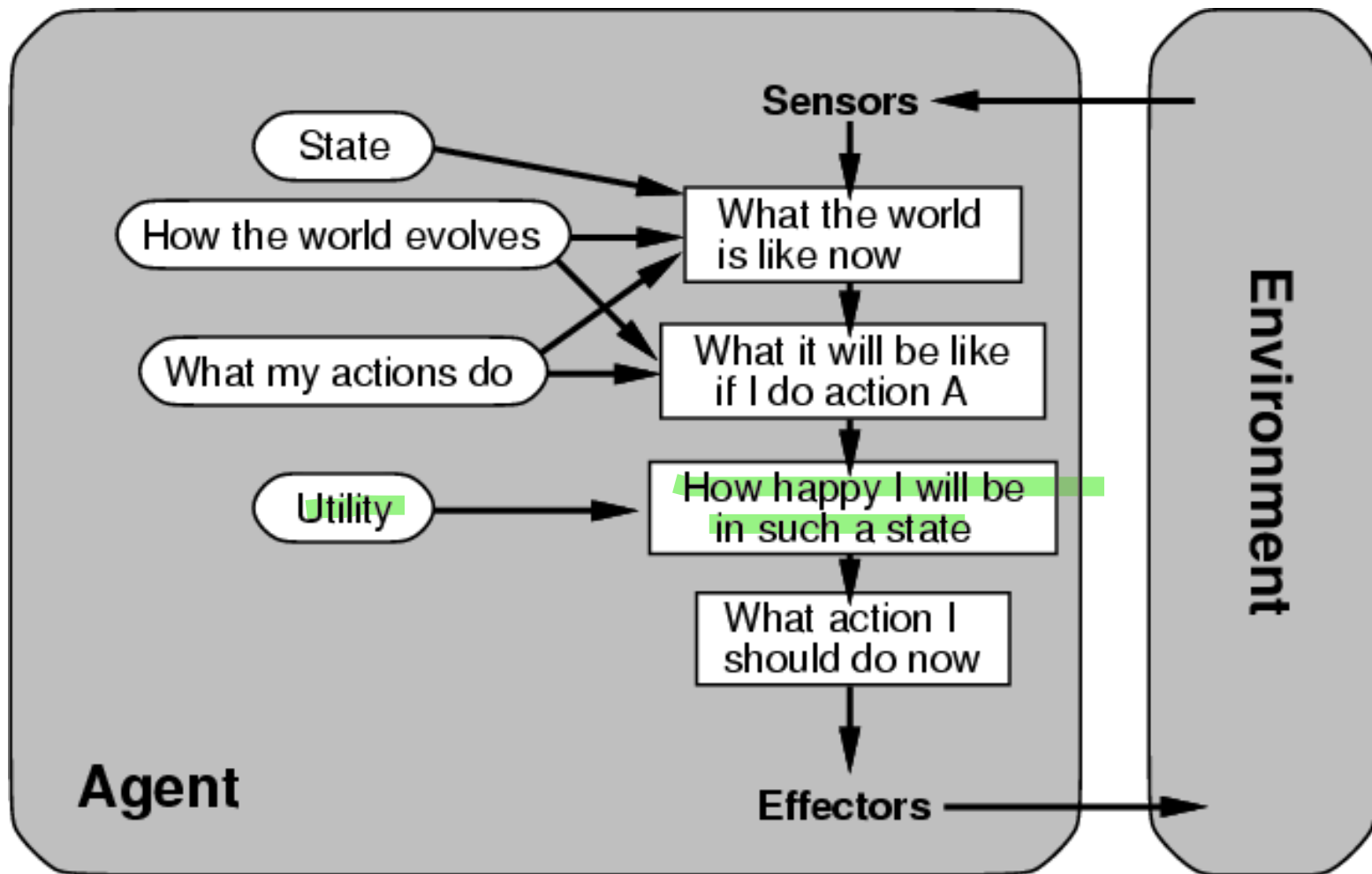


Goal-based agents

- Choose actions so as to achieve a (given or computed) goal.
- A goal is a description of a desirable situation.
- Keeping track of the current state is often not enough – need to add goals to decide which situations are good
- **Deliberative** instead of **reactive**.
- May have to consider long sequences of possible actions before deciding if goal is achieved – involves consideration of the future, *"what will happen if I do...?"*

Complete utility-based agent

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Utility-based agents

- When there are multiple possible alternatives, how to decide which one is best?
- A goal specifies a crude distinction between a happy and unhappy state, but often need a more general performance measure that describes “degree of happiness.”
- Utility function **$U: \text{State} \rightarrow \text{Reals}$** indicating a measure of success or happiness when at a given state.

Summary

- An **agent** perceives and acts in an environment, has an architecture, and is implemented by an agent program.
- An **ideal agent** always chooses the action which **maximizes its expected performance**, given its percept sequence so far.
- An **autonomous agent** **uses its own experience** rather than built-in knowledge of the environment by the designer.

Summary (Contd.)

- An **agent program** maps from percept to action and updates its internal state.
 - **Reflex agents** respond immediately to percepts.
 - **Goal-based agents** act in order to achieve their goal(s).
 - **Utility-based agents** maximize their own utility function.
- **Representing knowledge** is important for **successful agent design**.
- The **most challenging environments** are
 - **partially observable**
 - **stochastic**
 - **sequential**
 - **Dynamic**
 - **continuous**
 - **contain multiple intelligent agents**

